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OPTIMUM LOCATIONS OF SWITCHING CENTERS AND THE ABSOLUTE CENTERS AND MEDIANS OF A GRAPH*

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The concepts of the 'center' and the 'median vertex' of a graph are generalized to the 'absolute center' and the 'absolute median' of a weighted graph (a graph with weights attached to its vertices as well as to its branches). These results are used to find the optimum location of a 'switching center' in a communication network and to locate the best place to build a 'police station' in a highway system. It is shown that the optimum location of a switching center is always at a vertex of the communication network while the best location for the police station is not necessarily at an intersection. Procedures for finding these locations are given.

IN A communication network, such as a telephone interconnection system, there is usually a switching center S . All traffic flows (messages) within the network must arrive at the center S before they are processed and then sent to their proper destination. A model of such a system may be assumed to be a finite graph (network) G with weights (nonnegative numbers) attached to each of its vertices and branches. The weight w_i attached to the branch b_i of G represents the length (or the cost per unit capacity) of that element. The weight h_i attached to a vertex v_i of G represents the number of wires (lines) that must be connected between vertex v_i and the switching center S to handle the information flows that either originate or terminate at v_i . The problem is to find the exact location of the switching center S such that, say, the total length of the wires is a minimum. The problem of the optimum location of a police station (or a hospital) is somewhat similar. Consider a number of communities of different sizes that are interconnected by a highway system. We would like to find the location of the police station P such that the maximum distance from P is a minimum.

From the above discussion it is clear that the usual graph theoretic concepts of the center and the median cannot be used.^[1,2] Consider a

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graph G ; a point x on G is a point along some branch of G that may or may not be a vertex of G . The distance between any two points x and y on G , represented by $d(x, y)$, is the length of the shortest path in G between points x and y , where the length of a path is the sum of the weights of the branches of that path.^[3,4] We define a point x_0 on an element of a weighted n -vertex graph G to be an *absolute center* of G , if for every point x on G

$$\max_{1 \leq i \leq n} h_i d(v_i, x_0) \leq \max_{1 \leq i \leq n} h_i d(v_i, x). \quad (1)$$

Similarly, we define a point y_0 on G to be an *absolute median* of G , if for every point y on G

$$\sum_{i=1}^{i=n} h_i d(v_i, y_0) \leq \sum_{i=1}^{i=n} h_i d(v_i, y). \quad (2)$$

The absolute median of a graph may be identified with the optimum location of the switching center in a communication network, and the absolute center may be identified with the optimum location of the police station in a highway system.

CENTERS AND MEDIANS

A SQUARE matrix $D = [d_{ij}]$ of order n is a distance matrix of an n -vertex (nonoriented) graph G , if^[4]

$$d_{ij} = \begin{cases} d(v_i, v_j), & \text{for } i, j = 1, 2, \dots, n, \text{ and } i \neq j, \\ d(v_i, v_i) = 0, & \text{for } i = j, i = 1, 2, \dots, n. \end{cases} \quad (3)$$

Let d_i^m be the maximum entry in the i th column of D . If $d_c^m = \min(d_1^m, d_2^m, \dots, d_n^m)$, then v_c is a center of G . Similarly, the median G may be defined, in terms of D , as follows: Let d_i^s be the sum of the entries of the i th column of G , then if $d_b^s = \min(d_1^s, d_2^s, \dots, d_n^s)$, v_b is the median vertex of G .

Consider the graph of Fig. 1(a). The distance matrix of that graph is

$$D = \begin{vmatrix} 0 & 10 & 24 & 20 & 34 \\ 10 & 0 & 14 & 12 & 24 \\ 24 & 14 & 0 & 12 & 10 \\ 20 & 12 & 12 & 0 & 20 \\ 34 & 24 & 10 & 20 & 0 \end{vmatrix}. \quad (4)$$

It is easily determined that v_4 is the center of the graph of Fig. 1(a) and that v_2 and v_3 are two possible choices for the median of that graph. The distance of the vertex farthest away from the center of a graph is called the radius r_0 of G . In general, we may write

$$r_0 = \min_{1 \leq j \leq n} \max_{1 \leq i \leq n} d(v_i, v_j). \quad (5)$$

For example, the radius of the graph of Fig. 1(a) is 20. Let us pick a point x_0 on the branch $b(v_2, v_3)$ such that $d(v_2, x_0) = 6$, as shown in Fig. 1(b). It can be seen that for the graph of Fig. 1(b)

$$\max_{1 \leq i \leq 5} d(v_i, x_0) = 18 < d(v_4, v_1) = 20. \quad (6)$$

From the above discussion it follows that there are points on a graph, such as x_0 in the graph of Fig. 1(b), which, roughly speaking, are more centrally located than the center of G . The concept of the absolute median is a little more general, because we have also attached different weights to the vertices of the graph. Similarly, the concept of absolute median is a generalization of the concept of median. The rest of this

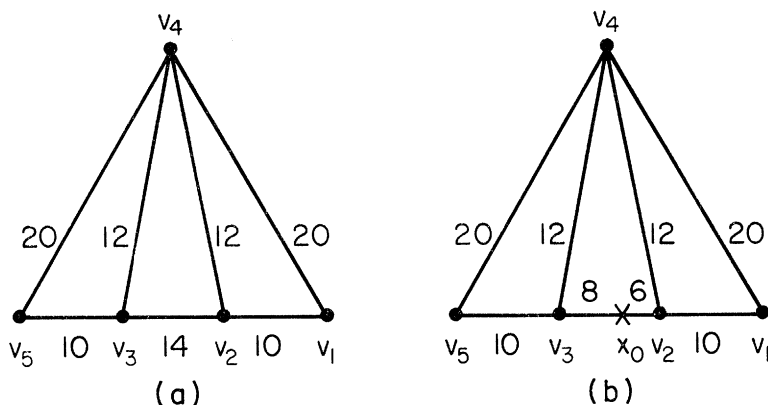


Fig. 1. (a) Center and median of a graph, and (b) the absolute center.

paper is devoted to describing methods for finding the absolute center and median of a graph.

THE ABSOLUTE CENTER AND THE RADIUS OF A GRAPH

THE ABSOLUTE center of a graph may be redefined as a point x_0 on G such that

$$\min_{x \text{ on } G} \max_{1 \leq i \leq n} h_i d(v_i, x) = \max_{1 \leq i \leq n} h_i d(v_i, x_0) = r(x_0), \quad (7)$$

where $r(x_0)$ is the absolute radius of G . Let the number of branches in G be m , then equation (7) may be written, in a more expanded form, as

$$\min_{1 \leq j \leq m} [\min_{x \text{ on } b_j} \max_{1 \leq i \leq n} h_i d(v_i, x)] = r(x_0). \quad (8)$$

However from equation (8), we can conclude that the problem of finding the absolute center x_0 may be reduced to the problem of solving m simpler 'min-max' problems. More precisely, let x_j be a point on b_j such that

$$\min_{x \text{ on } b_j} \max_{1 \leq i \leq n} h_i d(v_i, x) = \max_{1 \leq i \leq n} h_i d(v_i, x_j) = r(x_j). \quad (9)$$

Then, we call x_j a local center of G on b_j , and furthermore, we have

$$\min_{1 \leq j \leq n} [r(x_j)] = r(x_0). \quad (10)$$

Therefore, if the location of the local centers $x_1, x_2, \dots, x_m$ on branches

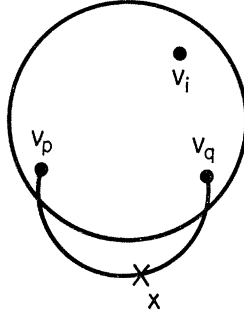


Fig. 2. An illustration of a distance $d(x, v_i)$.

$b_1, b_2, \dots, b_m$ are somehow determined, then the point x_0 is among the set of points $x_1, x_2, \dots, x_n$ and is the point in the set for which $r(x_j)$ is a minimum. We will now describe a method for locating the local center x_j on branch b_j of G .

$$\begin{aligned} a_i &= (1/2)[B_j + d(v_q, v_i) - d(v_p, v_i)] \\ d_i &= (1/2)[B_j + d(v_q, v_i) + d(v_p, v_i)] \end{aligned}$$

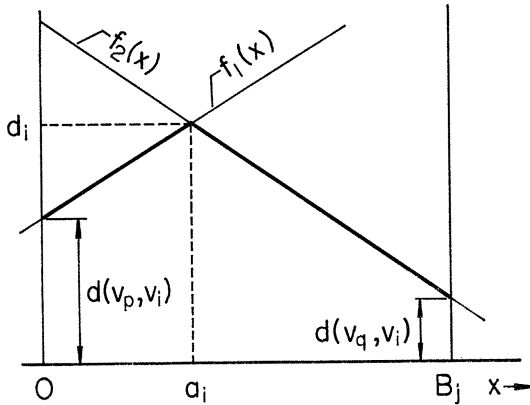


Fig. 3. A plot of $d(v_i, x)$.

Let us consider the graph of Fig. 2, and the branch $b_j(v_p, v_q)$ and an arbitrary vertex v_i in that graph. Let x be an arbitrary point on $b_j(v_p, v_q)$, then

$$d(v_i, x) = \min[d(x, v_p) + d(v_p, v_i), d(x, v_q) + d(v_q, v_i)]. \quad (11)$$

Let $d(v_p,v_q)=B_j$ and $d(v_p,x)=x$, then $d(v_q,x)=B_j-x$. In these terms, equation (11) may be written as

$$d(v_i,x)=\min[x+d(v_p,v_i),B_j-x+d(v_q,v_i)]. \tag{12}$$

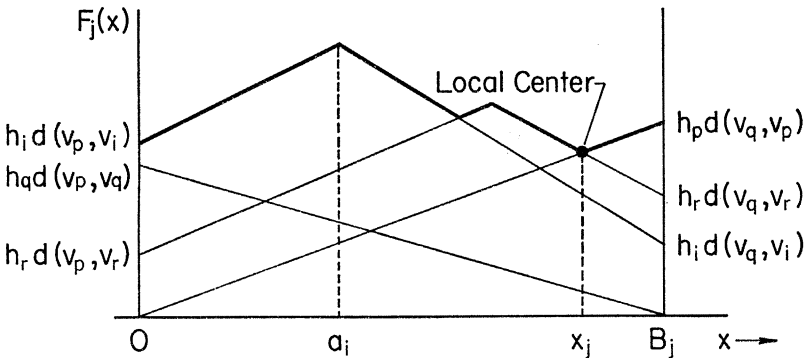


Fig. 4. A representative plot of $F_j(x)$.

Let $f_1(x)=x+d(v_p,v_i)$ and $f_2(x)=B_j-x+d(v_q,v_i)$. The functions $f_1(x)$ and $f_2(x)$ represent straight lines and, therefore, can be plotted easily. These plots are shown in Fig. 3. We would like to obtain a plot of $d(v_i,x)=\min[f_1(x),f_2(x)]$, in the interval $0\leq x\leq B_j$. Such a plot can be

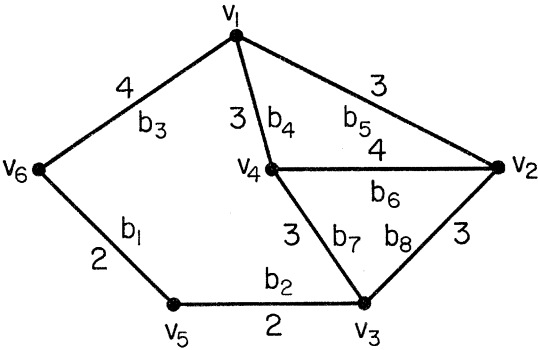


Fig. 5. An example to demonstrate the process for finding the absolute center.

easily obtained from the plots of $f_1(x)$ and $f_2(x)$ and is shown in Fig. 3 by the thick dark line. In Fig. 4, the function $h_i d(v_i,x)$ is plotted. Similarly the plots of other functions $h_1 d(v_1,x), h_2 d(v_2,x), \dots, h_n d(v_n,x)$ can be obtained in the interval $0\leq x\leq B_j$. These functions are all plotted on the same axes as shown in Fig. 4. To find the curve of the function

$$F_j(x)=\max_{1\leq i\leq n} h_i d(v_i,x), \quad \text{for } x \text{ on } b_j \tag{13}$$

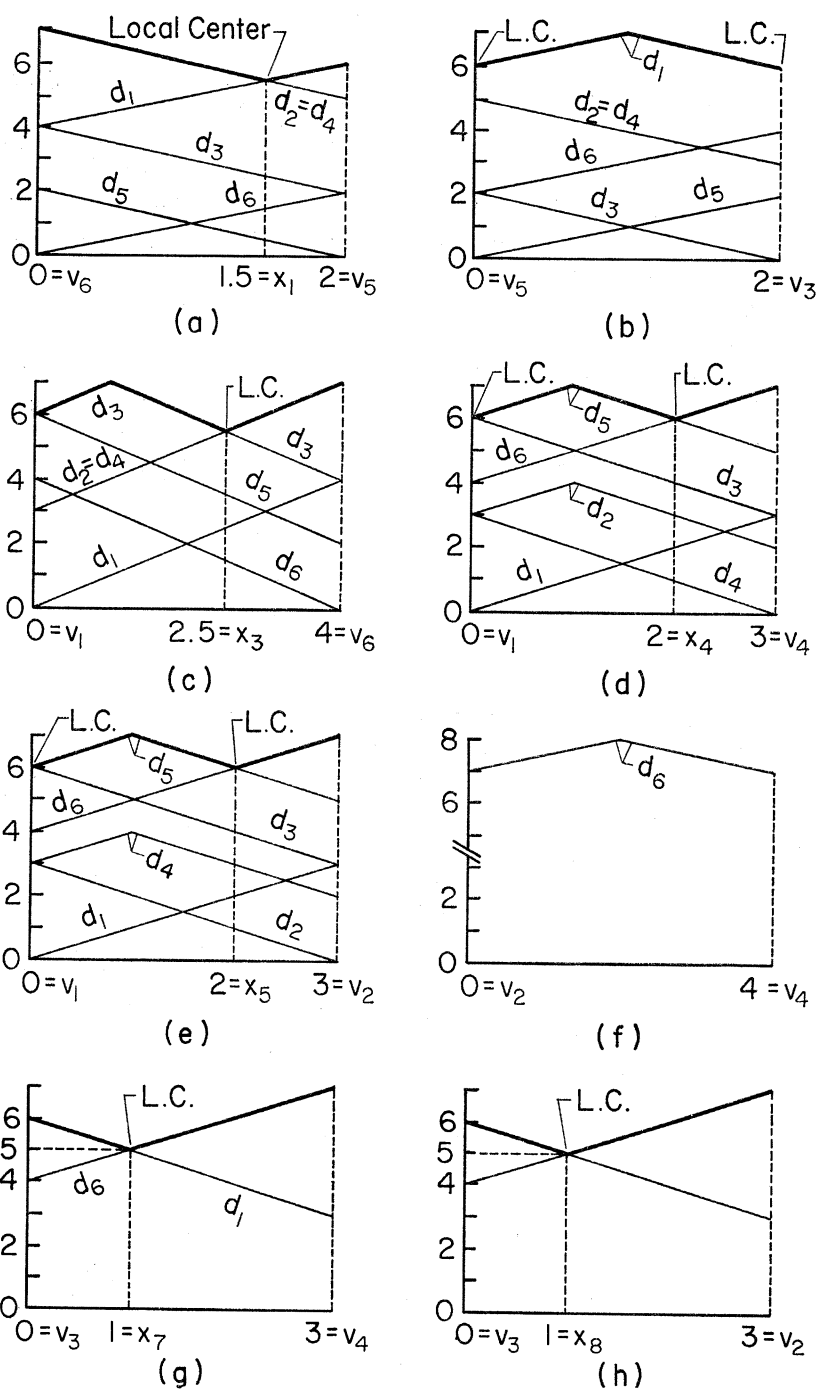


Fig. 6. Determination of local centers for a graph with 8 branches.

we must pick for each value of x , ($0 \leq x \leq B_j$), the maximum of all of the function plotted. The curve of the function $F_j(x)$ is also shown in Fig. 4 by the thick dark line. Now, to find the point x_j on $b_j(v_p, v_q)$ we must find the point at which $F_j(x)$ is a minimum. Since $F_j(x)$ is a 'piecewise linear function,' $F_j(s)$ has a finite number of local minima each of which can be easily computed; therefore, there is no difficulty in calculating the position of the point x_j .

The details of the above process may best be explained by a numerical example. In this example we will assume that weights attached to all vertices is equal to one. Clearly, this assumption does not have much effect on the details of the procedure. Example: Consider the graph G of Fig. 5. We would like to find the location of the absolute center of G . We first compute the distance matrix for graph G of Fig. 5.

$$D = \begin{vmatrix} 0 & 3 & 6 & 3 & 6 & 4 \\ 3 & 0 & 3 & 4 & 5 & 7 \\ 6 & 3 & 0 & 3 & 2 & 4 \\ 3 & 4 & 3 & 0 & 5 & 7 \\ 6 & 5 & 2 & 5 & 0 & 2 \\ 4 & 7 & 4 & 7 & 2 & 0 \end{vmatrix}. \quad (14)$$

The next step is to calculate the locations of the local centers on all branches of G . For the branch b_1 we compute $F_1(x)$, which is shown in Fig. 6(a). The location of local center x_1 is found to be at a point on b_1 such that $d(v_6, x_1) = 1.5$ and $F_1(1.5) = 5.5$. For branch b_2 , we can find that $F_2(x) \geq 6$ for all x on b_2 as shown in Fig. 6(b); therefore, it is not necessary to find the local center on b_2 . On branch b_3 , the function $F_3(x)$ is found to be minimum at $x_3, d(x_3, v_1) = 2.5$, and again $F(x_3) = 5.5$. For branches b_4 and b_5 , the functions $F_4(x)$ and $F_5(x) \geq 6$, see Fig. 6(d) and (e). Consider the branch b_6 . Since $d(v_6, v_4) = 7$ and $d(v_6, v_2) = 7$, we can immediately decide that $F_6(x) \geq 7$, for all x on b_6 . However, on branch b_7 there is a local center $x_7, d(x_7, v_3) = 1$, such that $F_7(x_7) = 5$. Also there is a point x_8 on branch b_8 such that $F_8(x_8) = 5$. Now, clearly $\min F_i(x_i) = F_7(x_7) = F_8(x_8)$; therefore, we have two absolute centers x_7 and x_8 and $\max_{1 \leq i \leq 8} d(v_i, x_7) = 5$, which means that the absolute radius of G is 5.

ABSOLUTE MEDIAN

THE FOLLOWING theorem is a generalization of a result communicated to the author by A. J. GOLDSTEIN,^[5] who has shown that an absolute median of a connected circuit-less graph (a tree) is always at a vertex.*

THEOREM. *An absolute median of a graph is always at a vertex of a graph.*

* Dr. Goldstein did not use the term absolute median or for that matter the term median.

Proof. To prove this theorem, we will show that if x_0 is an arbitrary point on G and $x_0 \neq v_i$ for $i = 1, 2, \dots, n$, then there always exists a vertex v_m in G such that

$$\sum_{i=1}^{i=n} h_i d(v_i, x_0) \geq \sum_{i=1}^{i=n} h_i d(v_i, v_m), \quad (15)$$

which will prove that a vertex of G will be an absolute median. Let x_0 be on a branch $b(v_p, v_q)$ of G . To begin with, we know that

$$d(v_i, x_0) = \min[d(x_0, v_p) + d(v_p, v_i), d(x_0, v_q) + d(v_q, v_i)]. \quad (16)$$

Let $i_1, i_2, \dots, i_n$ represent a rearrangement of the set of integers $1, 2, \dots, n$ such that

$$d(x_0, v_{i_k}) = d(x_0, v_p) + d(v_p, v_{i_k}), \quad \text{for } k = 1, 2, \dots, r, \quad (r \leq n) \quad (17)$$

$$\text{and } d(x_0, v_{i_k}) = d(x_0, v_q) + d(v_q, v_{i_k}), \quad \text{for } k = r+1, \dots, n. \quad (18)$$

Then we may write

$$\begin{aligned} \sum_{i=1}^{i=n} h_i d(v_i, x_0) &= \sum_{k=1}^{k=r} h_{i_k} [d(x_0, v_p) + d(v_p, v_{i_k})] \\ &\quad + \sum_{k=r+1}^{k=n} h_{i_k} [d(x_0, v_q) + d(v_q, v_{i_k})]. \end{aligned} \quad (19)$$

We will study equation (19) by considering the following two possible cases separately:

$$\text{Case (a): } \sum_{k=1}^{k=r} h_{i_k} \geq \sum_{k=r+1}^{k=n} h_{i_k}.$$

$$\text{Case (b): } \sum_{k=1}^{k=r} h_{i_k} < \sum_{k=r+1}^{k=n} h_{i_k}.$$

Case (a): We have $d(x_0, v_q) = d(v_p, v_q) - d(x_0, v_p)$, which when substituted in equation (19) results in

$$\begin{aligned} \sum_{i=1}^{i=n} h_i d(v_i, x_0) &= \sum_{k=1}^{k=r} h_{i_k} [d(x_0, v_p) + d(v_p, v_{i_k})] \\ &\quad + \sum_{k=r+1}^{k=n} h_{i_k} [d(v_p, v_q) - d(x_0, v_p) + d(v_q, v_{i_k})]. \end{aligned} \quad (20)$$

We also know that

$$d(v_q, v_p) + d(v_q, v_{i_k}) \geq d(v_p, v_{i_k}). \quad (21)$$

Making use of the inequality of equation (21), equation (20) can be changed to the following inequality

$$\begin{aligned} \sum_{i=1}^{i=n} h_i d(v_i, x_0) &\geq \sum_{k=1}^{k=r} h_{i_k} [d(x_0, v_p) + d(v_p, v_{i_k})] \\ &\quad + \sum_{k=r+1}^{k=n} h_{i_k} [d(v_p, v_{i_k}) - d(x_0, v_p)]. \end{aligned} \quad (22)$$

Rearranging the right-hand side of equation (22), we obtain

$$\begin{aligned} \sum_{i=1}^{i=n} h_i d(v_i, x_0) &\geq \sum_{i=1}^{i=n} h_i d(v_p, v_i) \\ &\quad + [\sum_{k=1}^{k=r} h_{i_k} - \sum_{k=r+1}^{k=n} h_{i_k}] d(x_0, v_p). \end{aligned} \quad (23)$$

From the hypothesis of *Case (a)*, we can conclude that

$$\sum_{i=1}^{i=n} h_i d(v_i, x_0) \geq \sum_{i=1}^{i=n} h_i d(v_p, v_i).$$

This however is the desired result.

Case (b): The proof of this case is very similar to the previous case. We substitute in equation (19), $d(x_0, v_p) = d(v_p, v_q) - d(v_q, x_0)$; then we obtain

$$\begin{aligned} \sum_{i=1}^{i=n} h_i d(v_i, x_0) &= \sum_{k=1}^{k=r} h_{i_k} [d(v_p, v_q) - d(v_q, x_0) + d(v_p, v_{i_k})] \\ &\quad + \sum_{k=r+1}^{k=n} h_{i_k} [d(x_0, v_q) + d(v_q, v_{i_k})]. \end{aligned} \quad (24)$$

However, equation (24), using entirely similar steps as in *Case (a)*, is reduced to the following inequality:

$$\begin{aligned} \sum_{i=1}^{i=n} h_i d(v_i, x_0) &\geq \sum_{i=1}^{i=n} h_i (v_i, v_q) \\ &\quad + [\sum_{k=r+1}^{k=n} h_{i_k} - \sum_{k=1}^{k=r} h_{i_k}] d(x_0, v_q). \end{aligned} \quad (25)$$

Making use of the hypothesis of *Case (b)*, we obtain:

$$\sum_{i=1}^{i=n} h_i d(v_i, x_0) > \sum_{i=1}^{i=n} h_i d(v_i, v_q) \quad (26)$$

which is what was desired, and hence the theorem.

The above theorem proves that although we can certainly define the term absolute median, such a point can never be any more of an absolute 'median' than a 'vertex median.'

CONCLUSIONS AND FURTHER PROBLEMS

WE HAVE shown that to find the optimum location of a switching center in a communication network, one can limit oneself to finding the vertex median of the corresponding graph. However, if one is concerned with building a hospital or a police station on a highway system in such a way that the distance of the farthest community from the police station is a minimum, then one must find the absolute center of the corresponding graph according to the procedure described in the third section. We may also look at the problem of the optimum location of the police station in the following way: Suppose h_i represents the average number of daily automobile accidents in community v_i , and furthermore let us assume that police from the station must visit each scene of an accident to make a report. With this sort of situation, the best place to build the station would be a median of the corresponding graph. The problem becomes more complicated if one is to find the optimum location of the police station with consideration given to a combination of both factors.

The problem of finding the location of the absolute center of a graph may be formulated in terms of game theory. To explain this, we will consider a problem that is slightly more general than the one considered

here. Let $d(x,y)$ be the distance between points x and y on G . The person X picks a point x on graph G . If person Y picks a point y on G , then X pays off $d(x,y)$ units of money to Y . X would like to pick a point from which the maximum distance is a minimum. Therefore the least possible amount of money that X can loose is

$$\min_{x \text{ on } G} \max_{y \text{ on } G} d(x,y) = r(x_0). \quad (27)$$

This game theoretic approach seems quite interesting, and the author is at the present studying the feasibility of such an approach.

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